



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO DAWOOD PUMPS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

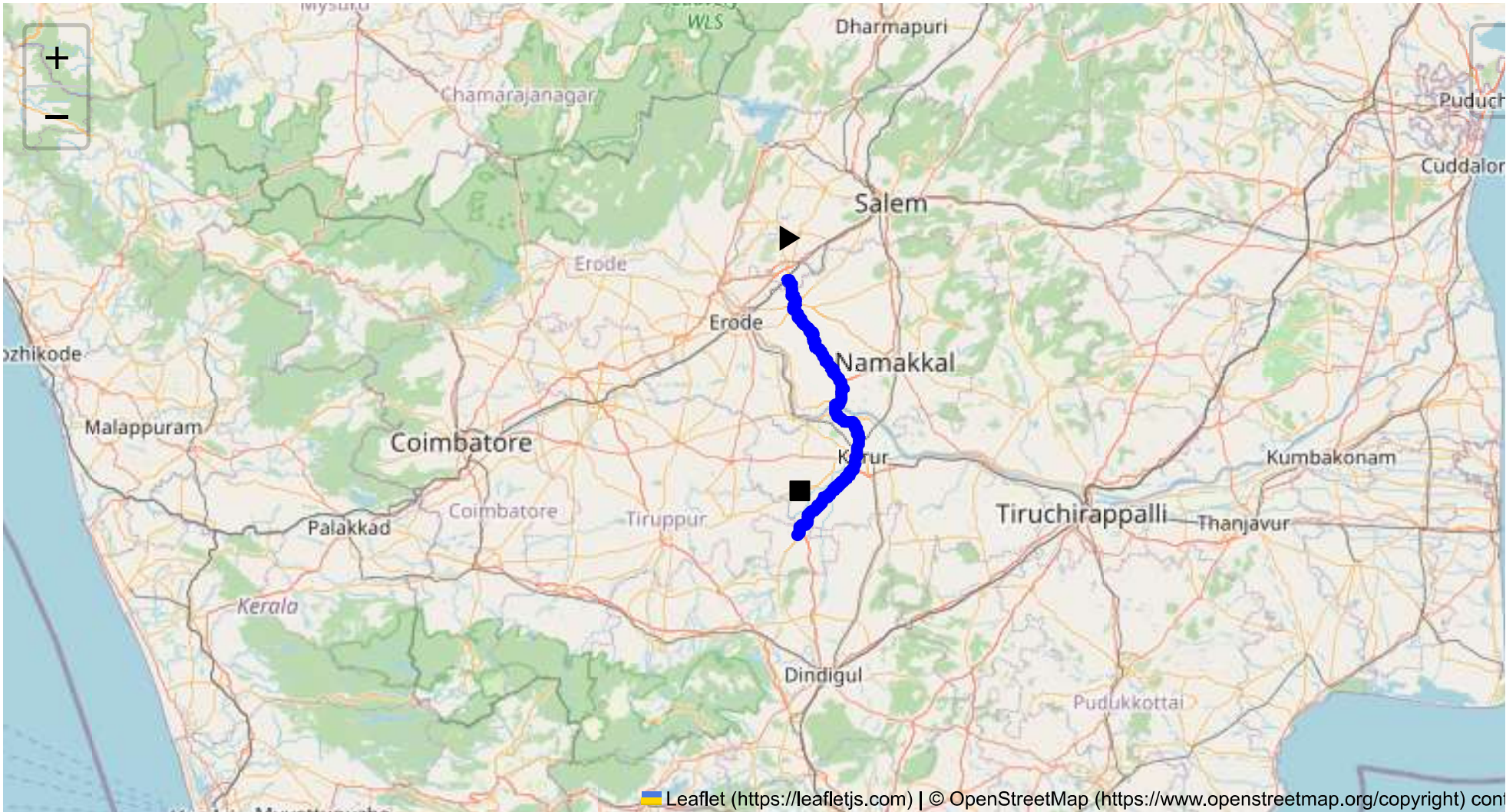
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 96.47 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (10.744696, 77.900578)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu, to PWV2+W5H, SH 193, Tamil Nadu, covers approximately 96.47 kilometers. The journey typically follows key highways that provide the most direct path, likely incorporating sections of SH 20 and SH 193, among others.

Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. Potential weather-related hazards include:

- **Monsoon Season** (June to September): Heavy rainfall can cause flooding and reduced visibility.
- **Summer Heat** (March to May): High temperatures could lead to road surface damage and driver fatigue.
- **Cyclone Risk** (October to December): Occasionally affects Tamil Nadu, potentially leading to route disruptions.

Traffic Patterns

- **Peak Hours:** Traffic is typically heavier during morning (8:00 AM to 10:00 AM) and evening (5:00 PM to 8:00 PM) due to local commuting.
- **Congestion-Prone Areas:** The towns and cities closest to SH 20 and SH 193 might experience congestion, particularly near markets or junctions.

Road Quality and Infrastructure

- **Highway Conditions:** Generally, the infrastructure is adequate for heavy vehicles, but some sections may have potholes or narrow lanes.
- **Rural Roads:** Off-highway roads may be poorly maintained and have limited facilities for heavy vehicles.

Alternative Routes for Emergencies

In emergencies, consider:

- Using local bypasses around major towns.
- Consulting digital maps for real-time alerts and alternative paths, especially during inclement weather.

Local Regulations for Hazardous Material Transport

- **Permit Requirements:** Ensure all necessary permits are secured.
- **Time Restrictions:** Check for any local restrictions on transit timings for hazardous materials.

Historical Incidents

- There could be historical data of incidents primarily in congested areas or during adverse weather conditions.
- Incidents often result from poor visibility, mechanical failure, or road quality.

Environmental Considerations

- **Sensitive Areas:** Monitor for potential pollution near water bodies or natural reserves.

- **Encroachment:** Be aware of agricultural activity near roadways, which may impact road conditions.

Communication Coverage

- **Dead Zones:** Rural areas between major towns may experience reduced mobile coverage.
- **GPS Reliability:** Map applications might have delayed updates in less populated areas.

Emergency Response Times

- **Urban Areas:** Approximately 15-30 minutes due to the presence of emergency services.
- **Rural Areas:** Could exceed 45 minutes to an hour depending on distance from nearest town.

Overall Summary of Risk Assessment

The route is manageable for heavy vehicles, though requires careful planning:

- Be vigilant during monsoon and cyclone seasons.
- Avoid peak congestion times and maintain steady speeds through less populated areas.
- Plan fuel stops and breaks efficiently to counter communication gaps and potential dead zones.
- Comply with local regulations for transporting hazardous materials.
- Engage with traffic updates and route advisories for alternative paths in emergencies.

Overall, ensure thorough checks and driver preparedness to mitigate risks that could arise from weather, road conditions, or traffic congestion. Regular communication with a home base or dispatch is recommended to provide situational awareness and support.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16755, 78.01568	30 KM/Hr
7	Turn	Medium	11.15579, 78.02169	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14485, 78.02282	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.10084, 78.00555	30 KM/Hr
11	Turn	High	10.77130, 77.91658	15 KM/Hr
12	Turn	High	10.77218, 77.91273	15 KM/Hr
13	Blind Spot	Blind Spot	10.77350, 77.90980	10 KM/Hr
14	Blind Spot	Blind Spot	10.74434, 77.90003	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
22	hospital	Government Hospital, Aravakurtchi	10.77208, 77.909648	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
21	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots